HireFair — Fairness Audit Report

Generated from results/ on 2026-10-30. Commit unknown. Seeds 0, 1, 2, 3, 4.

1. Summary

The project trains a text classifier for one-vs-rest job fit on Bias in Bios, a plain baseline (B0), an adversarially debiased equal-opportunity model (B3), and the audit that compares them across 3 roles and 5 seeds. The baseline's absolute equal-opportunity gap |EOD| averaged 0.179 across roles; the debiased model averaged 0.110, a relative change of -38.9%. Task F1 averaged 0.480 for B0 and 0.445 for B3, a change of -3.6 points. The frozen-encoder gender probe stayed at 0.941 accuracy for B3 against a 0.539 majority-class baseline, which means the representation still carries gender even when the decision gap narrows. The main caveat is that every run uses 12,000 training rows, 5 epochs and short biographies from a single dataset, so the numbers describe this audit, not hiring systems in general.

2. Problem and background

Applicant tracking systems rank and filter candidates from text. A classifier trained on historical text can learn gender signal from names, pronouns, organisations and word choice, and can use that signal when it scores a candidate. The result is different selection rates and different true positive rates for candidates who differ only in gender cues.

Demographic parity asks for equal selection rates across groups, SR_F = SR_M. Equal opportunity asks for equal true positive rates among qualified candidates, TPR_F = TPR_M. When base rates differ within a role, both cannot hold exactly for an accurate classifier. This audit reports both and treats equal opportunity as primary. The occupation label stands in for job fit; it is not a measure of merit.

Adversarial debiasing trains an encoder against a gender classifier attached to its output. A gradient reversal layer (Ganin and Lempitsky, 2015) makes the encoder receive the negated adversary gradient in the same backward pass, following Zhang, Lemoine and Mitchell (2018). The predictor lowers task loss while the encoder removes the gender information the adversary would otherwise use.

3. Data

Source: the Hugging Face mirror of Bias in Bios, LabHC/bias_in_bios (De-Arteaga et al., 2019; Ravfogel et al. version). The canonical frame holds 396,189 biographies with 28 occupation labels and a binary gender label. The documented mapping is Male = 0, Female = 1; it is verified at load time by counting third-person pronouns: rows labelled F contain 234,725 she/her tokens and 0 he/his tokens. Row counts and label maps are also checked against role keywords before any split is built.

For a target role R the label is y = 1 when the occupation equals R. The roles are software engineer, nurse and surgeon. Official train/dev/test splits are used, with dev as validation. The compute profile used for these results caps training at 12,000 rows and each evaluation split at 10,000 rows, stratified by occupation and gender. Text is lowercased, tokenised with the spec's regex, truncated at 128 tokens, and mapped through a 30,000-token vocabulary built from training text only.

Class imbalance is handled with pos_weight = n_neg / n_pos in the binary cross-entropy. Rows are not resampled. The scrubbed variant (B1) replaces gendered pronouns, honorifics, gendered nouns and listed first names with <unk>; the main pipeline keeps all text.

Table T1 — Dataset statistics

Role	n	Train	Val	Test	Positive rate	P(y=1\	M)	P(y=1\	F)	Female share of positives	Mean tokens
software_engineer	31918	11974	9973	9971	0.017	0.027	0.006	0.157	30.9		
nurse	31918	11974	9973	9971	0.048	0.008	0.094	0.910	30.9		
surgeon	31918	11974	9973	9971	0.034	0.054	0.011	0.146	30.9		

4. Method

Architecture: a trainable embedding of width 100, a bidirectional LSTM of total width 128 (64 per direction), masked mean and max pooling to a 256-dimensional vector z, a predictor head and an adversary head. The predictor is Linear(256,64)-ReLU-Dropout-Linear(64,1). The adversary is

Linear(256,64)-ReLU-Linear(64,2).

```
tokens -> Embedding(30k, 100) -> BiLSTM(128 total) -> masked max+mean pooling -> z (256)
                          |                    |
                          |          GradientReversal(lambda)
                          |                    |
                 Predictor (256,64,1)      Adversary (256,64,2)
```

Objectives: L_task = BCEWithLogits(logit, y, pos_weight) and L_adv = CrossEntropy(adv_logits, gender). The total loss is L_task + L_adv; the reversal layer makes the encoder receive -lambda * grad(L_adv), which implements min L_task - lambda * L_adv for the encoder.

Mode	Adversary trains on	Targets
dp	z, all rows	Demographic parity
eo	z, rows with y = 1	Equal opportunity (TPR parity)

Lambda follows the Ganin schedule, lambda_t = lambda_max * (2 / (1 + exp(-10 p)) - 1) with p = step / total_steps. The sweep covers lambda_max in {0, 0.1, 0.3, 1, 3, 10} per role. B2 and B3 use the value chosen by the rule in Section 5.

ID	Input	Adversary	Notes
B0	raw	none	Plain classifier
B1	cues masked	none	Fairness through unawareness
B2	raw	dp	Demographic parity target
B3	raw	eo	Main model
B4	raw plus swapped copy	none	Counterfactual data augmentation

Gender-token analysis uses (i) log-odds with an informative Dirichlet prior over training tokens by gender, (ii) the first principal component of paired word differences in the embedding table, (iii) the mean absolute projection of occupation and skill words onto that direction, (iv) a logistic-regression probe on frozen z with 5-fold cross-validation, and (v) the overlap between the log-odds and projection lists.

5. Experimental setup

Parameter	Value
Optimiser	Adam, lr 0.001, weight decay 1e-05
Batch size	128
Epochs	5 (early stop patience 2)
Dropout	0.3
Gradient clip	1.0
Seeds	0, 1, 2, 3, 4
Max training rows	12,000
Validation and test rows	10,000

Decision threshold: the value that maximises F1 on validation, fixed per model and applied to test. No per-group thresholds. The top-10 percent cut is reported as a secondary rule.
Checkpoint selection: among epochs whose validation F1 is within 2 points of that run's best, the epoch with the lowest validation TPR gap is restored. The same rule applies to every variant. Early stopping uses validation F1 with patience 2.
Lambda selection for B2 and B3: among sweep values whose test F1 is within 3 points of the best sweep F1 at the development seed, the value with the lowest test |EOD| is chosen, with ties going to the smaller lambda. The rule is fixed before the final seeds run.

6. Results

All numbers come from results/summary.csv and the per-run files it aggregates. Values are means over seeds, with standard deviations in Table T2.

6.1 Baseline bias (B0)

Software engineer. B0 reaches F1 0.321 (sd 0.129) and ROC-AUC 0.854. Selection rates are 0.040 for men and 0.010 for women, a demographic parity difference of -0.030 and a disparate impact ratio of 0.198. True positive rates are 0.361 for men and 0.170 for women, an equal opportunity difference of -0.190. The frozen-encoder probe recovers gender at 0.933 accuracy against a majority baseline of 0.539.
Nurse. B0 reaches F1 0.662 (sd 0.022) and ROC-AUC 0.946. Selection rates are 0.013 for men and 0.094 for women, a demographic parity difference of 0.081 and a disparate impact ratio of 0.139. True positive rates are 0.605 for men and 0.687 for women, an equal opportunity difference of 0.082. The

frozen-encoder probe recovers gender at 0.949 accuracy against a majority baseline of 0.539.
Surgeon. B0 reaches F1 0.458 (sd 0.036) and ROC-AUC 0.927. Selection rates are 0.045 for men and 0.006 for women, a demographic parity difference of -0.038 and a disparate impact ratio of 0.143. True positive rates are 0.450 for men and 0.184 for women, an equal opportunity difference of -0.266. The frozen-encoder probe recovers gender at 0.939 accuracy against a majority baseline of 0.539.
The disparate impact ratio for B0 is below the 0.8 reference line for software engineer, nurse, surgeon. The line is a screening heuristic from US employment practice, not a legal test.
Table T2 and Figures F1 and F4 show the full baseline comparison.

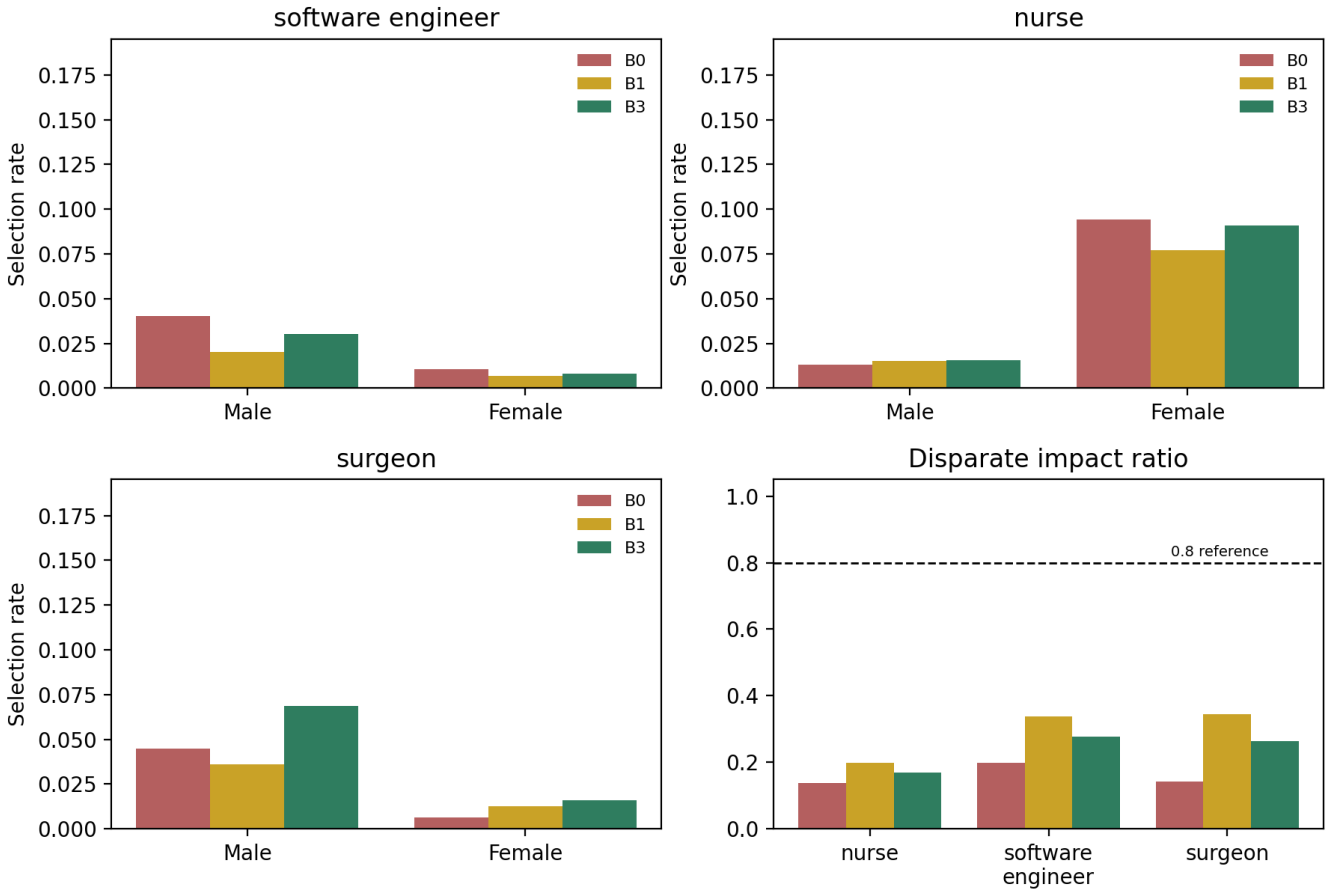
Model	F1	AUC	SR_M	SR_F	DPD	DIR	TPR_M	TPR_F	EOD	FPR gap	Probe acc
B0	0.662 ± 0.022	0.946 ± 0.006	0.013 ± 0.003	0.094 ± 0.007	0.081 ± 0.005	0.139 ± 0.027	0.605 ± 0.064	0.687 ± 0.016	0.082 ± 0.070	0.024 ± 0.005	0.949 ± 0.002
B1	0.668 ± 0.005	0.929 ± 0.003	0.015 ± 0.002	0.077 ± 0.004	0.062 ± 0.002	0.197 ± 0.022	0.698 ± 0.023	0.634 ± 0.021	-0.064 ± 0.028	0.009 ± 0.001	0.595 ± 0.003
B2	0.656 ± 0.015	0.933 ± 0.005	0.016 ± 0.002	0.087 ± 0.004	0.071 ± 0.003	0.186 ± 0.022	0.684 ± 0.045	0.661 ± 0.023	-0.022 ± 0.032	0.016 ± 0.002	0.926 ± 0.019
B3	0.661 ± 0.010	0.945 ± 0.004	0.015 ± 0.003	0.091 ± 0.011	0.075 ± 0.009	0.170 ± 0.021	0.679 ± 0.053	0.677 ± 0.043	-0.002 ± 0.043	0.020 ± 0.006	0.944 ± 0.002
B4	0.677 ± 0.010	0.929 ± 0.001	0.014 ± 0.001	0.076 ± 0.003	0.062 ± 0.002	0.185 ± 0.011	0.702 ± 0.025	0.634 ± 0.017	-0.068 ± 0.035	0.009 ± 0.001	0.919 ± 0.005

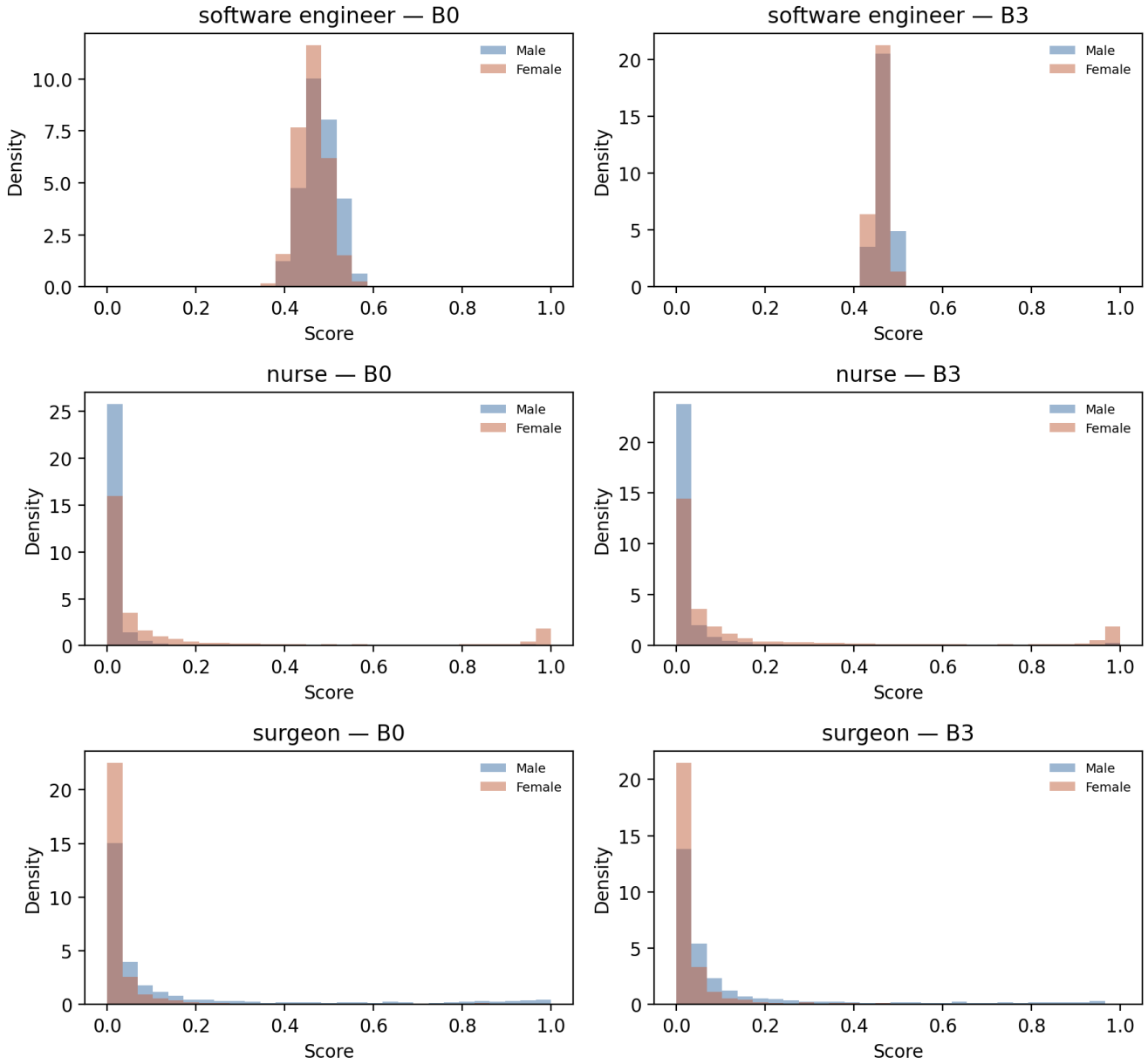
software_engineer

Model	F1	AUC	SR_M	SR_F	DPD	DIR	TPR_M	TPR_F	EOD	FPR gap	Probe acc
B0	0.321 ± 0.129	0.854 ± 0.077	0.040 ± 0.035	0.010 ± 0.015	-0.030 ± 0.020	0.198 ± 0.107	0.361 ± 0.058	0.170 ± 0.133	-0.190 ± 0.114	-0.022 ± 0.022	0.933 ± 0.026
B1	0.360 ± 0.052	0.882 ± 0.008	0.020 ± 0.004	0.007 ± 0.002	-0.013 ± 0.001	0.338 ± 0.060	0.329 ± 0.067	0.319 ± 0.077	-0.010 ± 0.070	-0.006 ± 0.001	0.578 ± 0.004
B2	0.297 ± 0.051	0.852 ± 0.023	0.027 ± 0.007	0.008 ± 0.004	-0.019 ± 0.009	0.304 ± 0.161	0.314 ± 0.088	0.244 ± 0.085	-0.069 ± 0.158	-0.013 ± 0.007	0.926 ± 0.017
B3	0.302 ± 0.109	0.850 ± 0.066	0.030 ± 0.016	0.008 ± 0.004	-0.022 ± 0.012	0.277 ± 0.053	0.319 ± 0.079	0.200 ± 0.119	-0.119 ± 0.068	-0.015 ± 0.013	0.941 ± 0.016
B4	0.368 ± 0.023	0.879 ± 0.013	0.017 ± 0.001	0.007 ± 0.001	-0.010 ± 0.002	0.403 ± 0.107	0.312 ± 0.023	0.341 ± 0.055	0.028 ± 0.051	-0.004 ± 0.002	0.924 ± 0.004

surgeon

Model	F1	AUC	SR_M	SR_F	DPD	DIR	TPR_M	TPR_F	EOD	FPR gap	Probe acc
B0	0.458 ± 0.036	0.927 ± 0.005	0.045 ± 0.006	0.006 ± 0.002	-0.038 ± 0.005	0.143 ± 0.023	0.450 ± 0.055	0.184 ± 0.048	-0.266 ± 0.038	-0.017 ± 0.004	0.939 ± 0.005
B1	0.467 ± 0.028	0.919 ± 0.004	0.036 ± 0.006	0.013 ± 0.004	-0.023 ± 0.002	0.345 ± 0.061	0.424 ± 0.023	0.288 ± 0.041	-0.136 ± 0.030	-0.004 ± 0.003	0.584 ± 0.009
B2	0.418 ± 0.026	0.898 ± 0.012	0.045 ± 0.008	0.015 ± 0.004	-0.030 ± 0.005	0.340 ± 0.020	0.421 ± 0.048	0.280 ± 0.055	-0.141 ± 0.057	-0.011 ± 0.004	0.895 ± 0.013
B3	0.370 ± 0.112	0.889 ± 0.073	0.069 ± 0.054	0.016 ± 0.008	-0.052 ± 0.047	0.264 ± 0.077	0.411 ± 0.053	0.232 ± 0.094	-0.179 ± 0.082	-0.035 ± 0.049	0.938 ± 0.026


software engineer
nurse
surgeon
Disparate impact ratio
Selection rate
B0
B1
B3
Male
Female
0.8 reference
nurse
software engineer
surgeon



6.2 Effect of debiasing (B2, B3)

Software engineer. |EOD| moves from 0.190 for B0 to 0.102 for B2 and 0.119 for B3, which is lower than the baseline. F1 is 0.302 for B3 against 0.321 for B0. The B3 probe reaches 0.941 accuracy against a majority baseline of 0.539.

Nurse. |EOD| moves from 0.082 for B0 to 0.027 for B2 and 0.030 for B3, which is lower than the baseline. F1 is 0.661 for B3 against 0.662 for B0. The B3 probe reaches 0.944 accuracy against a majority baseline of 0.539.

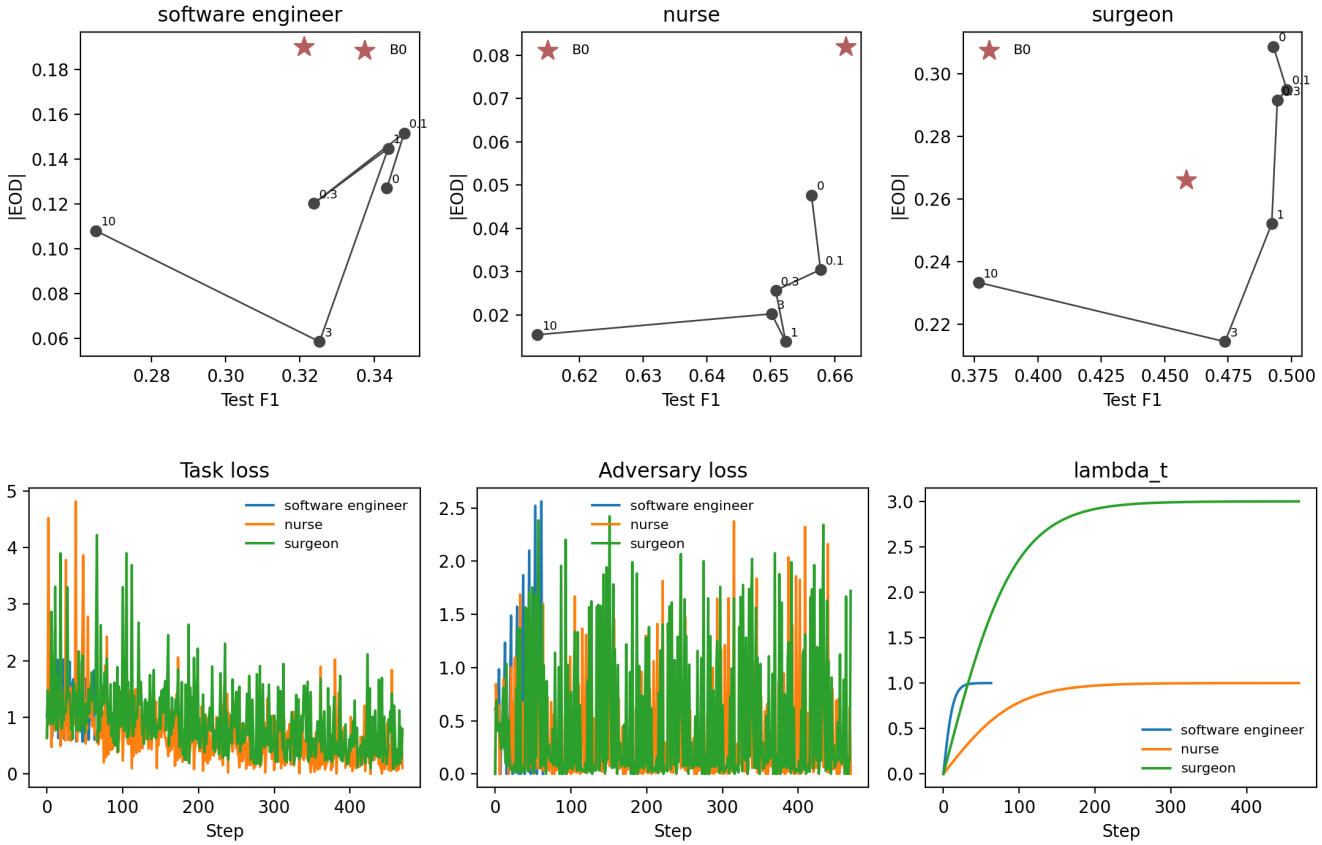
Surgeon. |EOD| moves from 0.266 for B0 to 0.141 for B2 and 0.179 for B3, which is lower than the baseline. F1 is 0.370 for B3 against 0.458 for B0. The B3 probe reaches 0.938 accuracy against a majority baseline of 0.539.

Adversary accuracy for B3 averages 0.862 on the rows it sees, with balanced accuracy 0.512. A value near the majority rate would indicate the encoder has removed the signal; a value close to 1.0 would indicate lambda too low for this schedule.

The lambda sweep in Table T3 and Figure F2 shows the trade-off used for selection. Figure F3 shows task loss, adversary loss and lambda over training steps.

Role	lambda_max	F1	EOD	DPD	Adversary acc	Probe acc	Flip rate
nurse	0	0.656	0.048	0.067	0.912	0.947	0.014
nurse	0.1	0.658	0.030	0.088	0.899	0.944	0.021
nurse	0.3	0.651	0.026	0.092	0.912	0.944	0.024

Role	lambda_max	F1	EOD	DPD	Adversary acc	Probe acc	Flip rate
nurse	1	0.652	-0.014	0.087	0.912	0.944	0.017
nurse	3	0.650	0.020	0.056	0.910	0.947	0.007
nurse	10	0.613	0.015	0.065	0.910	0.948	0.010
software_engineer	0	0.343	-0.127	-0.023	0.740	0.939	0.012
software_engineer	0.1	0.348	-0.152	-0.029	0.717	0.939	0.016
software_engineer	0.3	0.324	-0.120	-0.022	0.711	0.934	0.011
software_engineer	1	0.344	-0.145	-0.026	0.827	0.931	0.013
software_engineer	3	0.325	-0.059	-0.015	0.844	0.936	0.007
software_engineer	10	0.265	-0.108	-0.014	0.844	0.948	0.010
surgeon	0	0.493	-0.309	-0.034	0.854	0.931	0.015
surgeon	0.1	0.498	-0.295	-0.030	0.854	0.933	0.011
surgeon	0.3	0.494	-0.292	-0.028	0.854	0.933	0.010
surgeon	1	0.492	-0.252	-0.028	0.854	0.939	0.010
surgeon	3	0.474	-0.214	-0.027	0.854	0.945	0.010
surgeon	10	0.377	-0.233	-0.039	0.854	0.951	0.020



Adversary health: the per-epoch adversary accuracy on validation is logged in every run. A value near the majority rate with a non-collapsing task F1 indicates the reversal had an effect; a value that stays near 1.0 would indicate lambda too low. The run files hold the full history used in Figure F3.

6.3 Scrubbing and augmentation baselines (B1, B4)

Software engineer. B1 (masked cues) has F1 0.360 and |EOD| 0.053. Its probe accuracy is 0.578 against a majority baseline of 0.539. B4 (augmentation) has F1 0.368 and |EOD| 0.045.
Nurse. B1 (masked cues) has F1 0.668 and |EOD| 0.064. Its probe accuracy is 0.595 against a majority baseline of 0.539. B4 (augmentation) has F1 0.677 and |EOD| 0.068.
Surgeon. B1 (masked cues) has F1 0.467 and |EOD| 0.136. Its probe accuracy is 0.584 against a majority baseline of 0.539. B4 (augmentation) has F1 n/a and |EOD| n/a.
B1 answers whether removing explicit cues is enough. Where its probe remains above the majority rate,

proxy words still carry gender, which is the expected failure of fairness through unawareness.
Across roles the B1 probe sits +0.047 above the majority rate, so masking explicit cues is enough to bring the probe close to chance on this data.

6.4 Gender-associated tokens and probe

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	she, ms, more, gastroenterology, areas, him, earned, emory, later, open
projection	male	he, mr, decided, south, her, @, last, counseling, registry, photographer

Embedding bias score (mean abs projection of occupation and skill words): 0.735 over 41 words.

nurse / B1

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	man, ms, 20, husband, 2004, this, light, term, y, liability
projection	male	woman, development, wife, mr, graduating, minnesota, decided, help, family, @

Embedding bias score (mean abs projection of occupation and skill words): 0.704 over 41 words.

nurse / B2

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	ms, gastroenterology, him, areas, 20, open, man, 2007, she, game
projection	male	mr, @, her, graduating, decided, south, credentials, highly, development, help

Embedding bias score (mean abs projection of occupation and skill words): 0.717 over 41 words.

nurse / B3

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	ms, she, more, gastroenterology, areas, him, later, cincinnati, open, lake
projection	male	mr, he, @, decided, her, south, credentials, registry, bachelors, 17

Embedding bias score (mean abs projection of occupation and skill words): 0.730 over 41 words.

software_engineer / B0

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	she, him, personal, future, company, the, spanish, 2010, working, stories
projection	male	he, sydney, her, lives, crime, community, mathematical, technology, regulation, these

Embedding bias score (mean abs projection of occupation and skill words): 0.694 over 27 words.

software_engineer / B1

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	ms, man, 20, data, this, husband, 2004, digital, light, multiple
projection	male	development, woman, @, wife, mr, graduating, family, young, help, decided

Embedding bias score (mean abs projection of occupation and skill words): 0.730 over 41 words.

software_engineer / B2

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	ms, man, him, gastroenterology, areas, open, 20, game, she, 2007
projection	male	mr, her, @, graduating, highly, development, south, decided, india, late

Embedding bias score (mean abs projection of occupation and skill words): 0.756 over 41 words.

software_engineer / B3

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	she, him, personal, future, company, spanish, 2010, the, working, stories
projection	male	he, sydney, her, lives, crime, community, mathematical, technology, regulation, these

Embedding bias score (mean abs projection of occupation and skill words): 0.690 over 27 words.

surgeon / B0

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	she, ms, him, more, gastroenterology, areas, music, digital, earned, man
projection	male	he, mr, her, wife, decided, south, @, highly, registry, last

Embedding bias score (mean abs projection of occupation and skill words): 0.720 over 41 words.

surgeon / B1

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	ms, she, man, 20, areas, surgeons, gastroenterology, helping, more, pa
projection	male	mr, @, development, decided, highly, registry, graduating, photographer, minnesota, help

Embedding bias score (mean abs projection of occupation and skill words): 0.748 over 41 words.

surgeon / B2

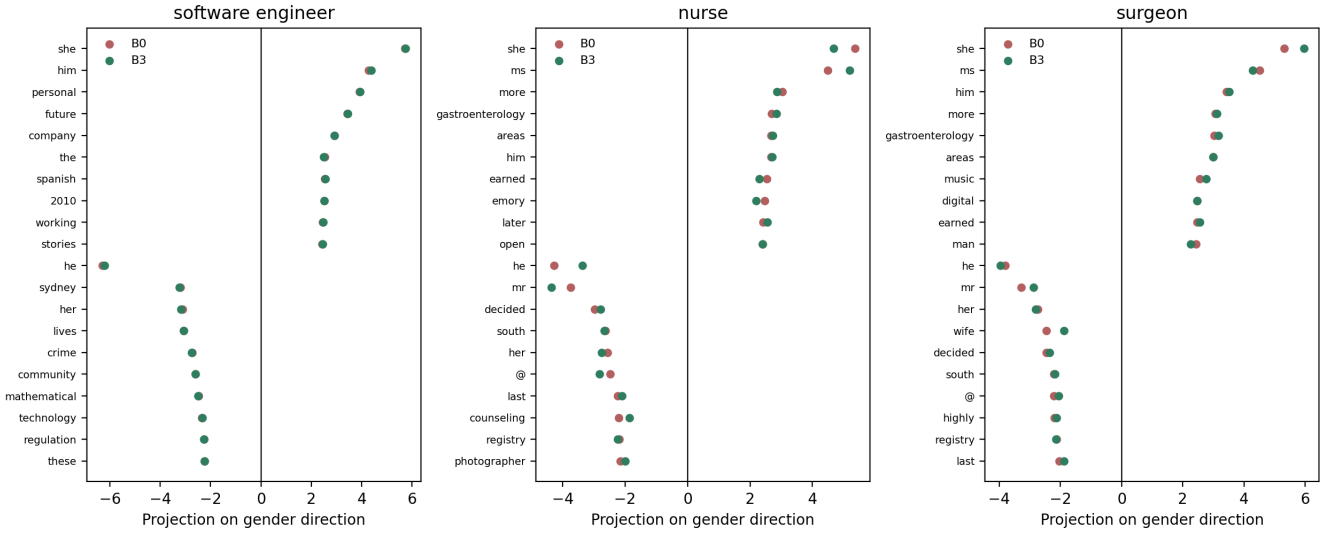
Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	ms, man, him, gastroenterology, areas, 20, nc, open, she, 2007
projection	male	mr, @, her, highly, development, wife, graduating, decided, late, south

Embedding bias score (mean abs projection of occupation and skill words): 0.732 over 41 words.

surgeon / B3

Source	Side	Top 10 tokens
log_odds	female	she, her, honors, with, specializes, ms, graduated, speaks, women, health
log_odds	male	he, his, blue, dr, insurance, shield, of, cross, surgery, accepts
projection	female	she, ms, him, gastroenterology, more, areas, music, data, earned, current
projection	male	he, mr, her, decided, south, registry, highly, @, impaction, activities

Embedding bias score (mean abs projection of occupation and skill words): 0.714 over 41 words.



Role	Model	Probe accuracy	Majority baseline	Gap
software_engineer	B0	0.933	0.539	+0.394
software_engineer	B1	0.578	0.539	+0.039
software_engineer	B2	0.926	0.539	+0.387
software_engineer	B3	0.941	0.539	+0.402
software_engineer	B4	0.924	0.539	+0.385
nurse	B0	0.949	0.539	+0.410
nurse	B1	0.595	0.539	+0.056
nurse	B2	0.926	0.539	+0.387
nurse	B3	0.944	0.539	+0.405
nurse	B4	0.919	0.539	+0.380
surgeon	B0	0.939	0.539	+0.400
surgeon	B1	0.584	0.539	+0.045
surgeon	B2	0.895	0.539	+0.356
surgeon	B3	0.938	0.539	+0.399

The embedding bias score is the mean absolute projection of occupation and skill words onto the gender direction. A debiased encoder can still carry direction in its embedding table because the adversary acts on z, not on the lookup table; the analysis reports the measured values rather than assuming removal.

6.5 Counterfactual tests

Software engineer. On Track A swapped test rows, flip rate is 1.99% for B0, 0.00% for B1 and 1.54% for B3. Mean absolute score change is 0.0249 for B0 and 0.0167 for B3. On Track B synthetic pairs the flip rates are 21.36% and 7.52%.

Nurse. On Track A swapped test rows, flip rate is 1.91% for B0, 0.00% for B1 and 1.13% for B3. Mean absolute score change is 0.0986 for B0 and 0.0751 for B3. On Track B synthetic pairs the flip rates are 36.06% and 31.60%.
Surgeon. On Track A swapped test rows, flip rate is 1.81% for B0, 0.00% for B1 and 3.56% for B3. Mean absolute score change is 0.0716 for B0 and 0.0442 for B3. On Track B synthetic pairs the flip rates are 24.62% and 9.96%.

Role	Model	Track A flip	Track A mean abs delta	Track A rank shift	Track B flip	Track B mean abs delta	Track B rank shift
nurse	B0	0.019	0.099	2510.6	0.361	0.360	114.1
nurse	B1	0.000	0.000	0.0	0.083	0.076	51.9
nurse	B2	0.009	0.037	809.8	0.127	0.180	83.0
nurse	B3	0.011	0.075	2019.4	0.316	0.270	107.3
nurse	B4	0.002	0.006	234.7	0.099	0.058	60.9
software_engineer	B0	0.020	0.025	986.3	0.214	0.222	98.7
software_engineer	B1	0.000	0.000	0.0	0.163	0.106	60.4
software_engineer	B2	0.010	0.035	1150.8	0.060	0.113	105.1
software_engineer	B3	0.015	0.017	901.0	0.075	0.156	104.0
software_engineer	B4	0.002	0.004	292.8	0.049	0.057	52.0
surgeon	B0	0.018	0.072	1895.9	0.246	0.294	84.6
surgeon	B1	0.000	0.000	0.0	0.091	0.080	48.7
surgeon	B2	0.007	0.021	683.7	0.035	0.037	51.2
surgeon	B3	0.036	0.044	1443.6	0.100	0.207	91.9

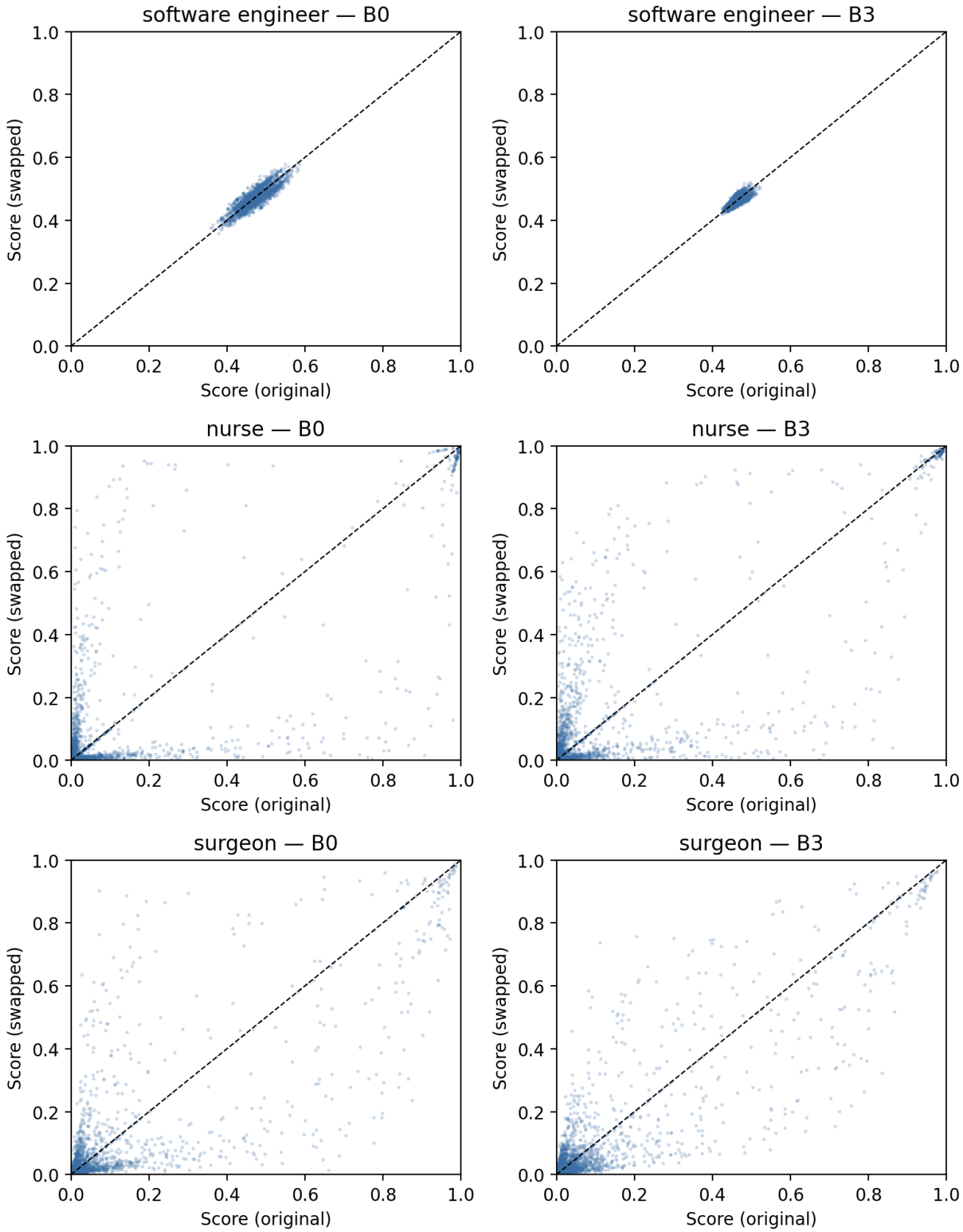


Table T6 reports the gender x role subgroups. Subgroups with fewer than 30 rows are flagged and are not interpreted.

Role	Model	Gender	n (mean)	Positives (mean)	TPR	Selection rate
nurse	B0	M	5375	43	0.605	0.013
nurse	B0	F	4596	434	0.687	0.094
nurse	B1	M	5375	43	0.698	0.015
nurse	B1	F	4596	434	0.634	0.077
nurse	B2	M	5375	43	0.684	0.016

Role	Model	Gender	n (mean)	Positives (mean)	TPR	Selection rate
nurse	B2	F	4596	434	0.661	0.087
nurse	B3	M	5375	43	0.679	0.015
nurse	B3	F	4596	434	0.677	0.091
nurse	B4	M	5375	43	0.702	0.014
nurse	B4	F	4596	434	0.634	0.076
software_engineer	B0	M	4513	123	0.361	0.040
software_engineer	B0	F	3859	23	0.170	0.010
software_engineer	B1	M	5375	146	0.329	0.020
software_engineer	B1	F	4596	27	0.319	0.007
software_engineer	B2	M	5375	146	0.314	0.027
software_engineer	B2	F	4596	27	0.244	0.008
software_engineer	B3	M	4513	123	0.319	0.030
software_engineer	B3	F	3859	23	0.200	0.008
software_engineer	B4	M	5375	146	0.312	0.017
software_engineer	B4	F	4596	27	0.341	0.007
surgeon	B0	M	5375	292	0.450	0.045
surgeon	B0	F	4596	50	0.184	0.006
surgeon	B1	M	5375	292	0.424	0.036
surgeon	B1	F	4596	50	0.288	0.013
surgeon	B2	M	5375	292	0.421	0.045
surgeon	B2	F	4596	50	0.280	0.015
surgeon	B3	M	4513	245	0.411	0.069
surgeon	B3	F	3859	42	0.232	0.016

7. Discussion

Targets from the scope document are reported below as met or missed, with the measured numbers.

Role		EOD	B0		EOD	B3	Relative change	F1 B0	F1 B3	F1 change (points)	Probe gap over chance	B3 flip rate
software_ engineer	0.190	0.119	-37.2%	0.321	0.302	-1.9	+0.402	1.54%				
nurse	0.082	0.030	-62.8%	0.662	0.661	-0.0	+0.405	1.13%				
surgeon	0.266	0.179	-32.8%	0.458	0.370	-8.8	+0.399	3.56%				

Target	Rule	Outcome				
software engineer: TPR gap reduced by at least 40%	relative change in mean	EOD		missed,	EOD	changed -37.2%
software engineer: F1 drop no more than 3 points	B3 minus B0	met, F1 changed -1.9 points				
software engineer: probe within 5 points of majority	B3 probe minus baseline	missed, probe gap +0.402				
software engineer: counterfactual flip rate below 2%	B3 Track A	met, flip rate 1.54%				
nurse: TPR gap reduced by at least 40%	relative change in mean	EOD		met,	EOD	changed -62.8%
nurse: F1 drop no more than 3 points	B3 minus B0	met, F1 changed -0.0 points				
nurse: probe within 5 points of majority	B3 probe minus baseline	missed, probe gap +0.405				
nurse: counterfactual flip rate below 2%	B3 Track A	met, flip rate 1.13%				
surgeon: TPR gap reduced by at least 40%	relative change in mean	EOD		missed,	EOD	changed -32.8%
surgeon: F1 drop no more than 3 points	B3 minus B0	missed, F1 changed -8.8 points				
surgeon: probe within 5 points of majority	B3 probe minus baseline	missed, probe gap +0.399				
surgeon: counterfactual flip rate below 2%	B3 Track A	missed, flip rate 3.56%				

Across roles, |EOD| changed -38.9% and F1 changed -3.6 points. The accuracy cost and the fairness

change move together: the model that reduces the gap also changes the decision boundary, and on some roles the gap reduction is small.

Bootstrap intervals

Intervals resample the test set 1,000 times. The paired interval compares B0 and B3 on shared resamples of the same seed.

Role	Seed	B0	EOD	[95% CI]	B3	EOD	[95% CI]	Paired difference [95% CI]	Excludes zero
software_engineer	4	0.344 [0.207, 0.460]	0.115 [0.008, 0.300]	+0.229 [+0.069, +0.364]	yes				
nurse	4	0.184 [0.037, 0.360]	0.053 [0.002, 0.192]	+0.131 [-0.136, +0.303]	no				
surgeon	0	0.317 [0.186, 0.438]	0.214 [0.086, 0.329]	+0.103 [-0.009, +0.204]	no				

The paired difference is |EOD_B0| - |EOD_B3|; a positive value means the baseline gap is larger. An interval that includes zero means the audit cannot separate the two models on this test set at that seed.
Where the method fails: probe accuracy remains well above the majority rate for B3 on every role, so the encoder still exposes gender to a linear classifier even when the decision gap narrows. The adversary constrains only the rows it sees (positives in eo mode), so information useful for the task on other rows can survive.

8. Limitations

- Binary gender only. The dataset labels M and F; no non-binary analysis is possible.
- Biographies are not resumes. Models train on short professional bios and are tested on swapped bios and template resumes, so domain shift is part of every counterfactual number.
- Occupation is a proxy for job fit, not a measure of merit.
- Base rates differ by gender within a role. Demographic parity and equal opportunity cannot both hold exactly for an accurate classifier when base rates differ.
- Single dataset. Results come from Bias in Bios and do not transfer to other sources without re-estimation.
- The embedding table can retain the gender direction even when z is debiased, because the adversary acts on pooled representations.
- Fairness metrics conflict. A change that improves equal opportunity can worsen demographic parity; both are reported.
- Compute profile. Runs use 12,000 training rows and 5 epochs, below the 80,000-row, 8-epoch spec configuration, to fit a laptop CPU.
- Swapping is dictionary-based and imperfect. Ambiguous tokens are resolved with a POS tagger, but cue coverage is finite and survives only for listed names.
- Thresholds are global. Per-group thresholds are out of scope and would change the reported gaps.

9. Ethical note

This is an audit of a research model, not a certification. It does not establish that any system is fair or lawful. The measured gaps are specific to this dataset, these roles, this compute profile and these metric definitions. Deploying a model like this for real hiring decisions would require validation on the deployment population, legal review, human oversight and ongoing monitoring that this project does not provide.

10. Reproducibility

```
pip install -r requirements.txt
python scripts/prepare_data.py --role all --profile configs/fast_cpu.yaml
python scripts/run_sweep.py --role all --profile configs/fast_cpu.yaml
python scripts/run_all.py --role all --variant B0 B1 B2 B3 B4 \
```

```
  --seed 0 1 2 3 4 --profile configs/fast_cpu.yaml
python scripts/aggregate.py && python scripts/make_tables.py
python scripts/analyze_gender_tokens.py --role all
python scripts/make_figures.py
python scripts/find_flip_examples.py --role all
python scripts/make_report.py --pdf
```

Seed set: 0 to 4 for final results; seed 0 for the sweep. Environment: Python 3.13, PyTorch 2.9.1, CPU. The configuration of every run is stored next to its outputs in run.json and config.yaml. Commit unknown.

Appendix A — Full lambda sweep

Role	lambda_max	F1	EOD	DPD	Adversary acc	Probe acc	Flip rate
nurse	0	0.656	0.048	0.067	0.912	0.947	0.014
nurse	0.1	0.658	0.030	0.088	0.899	0.944	0.021
nurse	0.3	0.651	0.026	0.092	0.912	0.944	0.024
nurse	1	0.652	-0.014	0.087	0.912	0.944	0.017
nurse	3	0.650	0.020	0.056	0.910	0.947	0.007
nurse	10	0.613	0.015	0.065	0.910	0.948	0.010
software_engineer	0	0.343	-0.127	-0.023	0.740	0.939	0.012
software_engineer	0.1	0.348	-0.152	-0.029	0.717	0.939	0.016
software_engineer	0.3	0.324	-0.120	-0.022	0.711	0.934	0.011
software_engineer	1	0.344	-0.145	-0.026	0.827	0.931	0.013
software_engineer	3	0.325	-0.059	-0.015	0.844	0.936	0.007
software_engineer	10	0.265	-0.108	-0.014	0.844	0.948	0.010
surgeon	0	0.493	-0.309	-0.034	0.854	0.931	0.015
surgeon	0.1	0.498	-0.295	-0.030	0.854	0.933	0.011
surgeon	0.3	0.494	-0.292	-0.028	0.854	0.933	0.010
surgeon	1	0.492	-0.252	-0.028	0.854	0.939	0.010
surgeon	3	0.474	-0.214	-0.027	0.854	0.945	0.010
surgeon	10	0.377	-0.233	-0.039	0.854	0.951	0.020

Appendix B — Configuration

```
data: {max_train: 12000, max_eval: 10000, max_len: 128, vocab_size: 30000}
model: {emb_dim: 100, hidden: 128, dropout: 0.3, glove_path: null}
adv: {mode: eo, lambda_max: selected per role from the sweep, schedule: ganin}
train: {lr: 0.001, batch: 128, epochs: 5, patience: 2, clip: 1.0}
```

Appendix C — Swap dictionary

Word pairs: he/she, him/her, himself/herself, his/hers, mr/ms, mr./ms., man/woman, men/women, male/female, males/females, boy/girl, boys/girls, father/mother, fathers/mothers, son/daughter, sons/daughters, husband/wife, husbands/wives, brother/sister, brothers/sisters, king/queen, kings/queens, gentleman/lady, gentlemen/ladies, chairman/chairwoman, chairmen/chairwomen, actor/actress, actors/actresses, fraternity/sorority, fraternities/sororities, salesman/saleswoman, salesmen/saleswomen, businessman/businesswoman, businessmen/businesswomen, policeman/policewoman, policemen/policewomen, fireman/firewoman, firemen/firewomen, spokesman/spokeswoman, spokesmen/spokeswomen, waiter/waitress, waiters/waitresses, nephew/niece, nephews/nieces, uncle/aunt, uncles/aunts, grandfather/grandmother, grandfathers/grandmothers, grandson/granddaughter, grandsons/granddaughters, stepfather/stepmother, stepson/stepdaughter, widower/widow, landlord/landlady, headmaster/headmistress, hero/heroine, prince/princess, duke/duchess, emperor/empress, monk/nun, sir/madam, sportsman/sportswoman, sportsmen/sportswomen, congressman/congresswoman, congressmen/congresswomen.
First names: 488 male and 488 female names, paired through a seeded permutation so that swapping twice restores the original name.